MATH 3312, SPRING 2026: HOMEWORK 10

(1) Let R be a PID and let $q_1,\dots,q_r \in R$ be irreducible elements generating *distinct* prime ideals. Suppose that, for $i = 1,\dots,r$, we have integers $m_i, n_{i,j} \geq 0$ for $j \leq m_i$ and set $M = \prod_{i=1}^r \prod_{j=1}^{m_i} [R/(q_i^j)]^{n_{i,j}}$.

(a) For any irreducible element $q \in R$ and any $n \geq 1$, give a formula for $\mathrm{length}(q^n \cdot M)$ in terms of the product decomposition.

Solution.

Observation 1. If $M = M_1 \times M_2$ is a direct product of R-modules and M_1 and M_2 have finite length, then so does M and we have $\mathrm{length}(M) = \mathrm{length}(M_1) + \mathrm{length}(M_2)$.

Proof. This follows from Problem 2 on Homework 9 applied to the submodule $N = M_1 \times \{0\} \leq M$ with $M/N \simeq M_2$. $\square$

Using this observation and Problem 1 on Homework 9, we see that

$$\mathrm{length}(q^n \cdot M) = \sum_{i=1}^{r} \sum_{j=1}^{m_i} n_{i,j} \mathrm{length}\left(R/(q_i^j/\gcd(q_i^j, q^n))\right)$$

If $(q) \neq (q_i)$, then the two irreducible elements are relatively prime, and the gcd is 1. In this case, we have

$$\mathrm{length}\left(R/(q_i^j/\gcd(q_i^j, q^n))\right) = \mathrm{length}\left(R/(q_i^j)\right) = j$$

by Problem 3 on Homework 9.

If $(q) = (q_i)$, then by the same problem, we get

$$\mathrm{length}\left(R/(q_i^j/\gcd(q_i^j, q^n))\right) = \max(0, j-n).$$

(b) Conclude that the prime ideals $(q_1),\dots,(q_r)$ and the integers m_i and $n_{i,j}$ depend on M only up to isomorphism.

Solution. Set $f(q,n) = \mathrm{length}(M) - \mathrm{length}(q^n \cdot M)$: this is a quantity *intrinsically* associated with M.

By the formula from part (a), we find:

$$f(q,n) = \begin{cases} \sum_{j=1}^{m_i} n_{i,j}[j - \max(0, j-n)] = \sum_{j=1}^{m_i} n_{i,j} \min(j,n) & \text{if } (q) = (q_i) \text{ for some } i \\ 0 & \text{otherwise.} \end{cases}$$

Therefore, the ideals $(q_1),\dots,(q_r)$ are determined uniquely as those generated by irreducible elements q with $f(q,n)$ *non-zero* for some $n \geq 0$.

Now, fix some i, and set $m = m_i$ and $n_j = n_{i,j}$. We want to know now that the quantities $f(q_i,n) = \sum_{j=1}^{m} n_j \min(j,n)$ for varying n determine both m and the coefficients n_j.

The integer m is the smallest integer for which we have $f(q_i,m) = f(q_i,m+1) = \sum_{j=1}^{m} n_j j$.

Given this, the coefficients n_j satisfy the matrix equation over $\mathbb{Z}$:

$$\begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 2 & \cdots & 2 \\ 1 & 2 & 3 & \cdots & 3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 2 & 3 & \cdots & m \end{bmatrix} \cdot \begin{bmatrix} n_1 \\ n_2 \\ n_3 \\ \vdots \\ n_m \end{bmatrix} = \begin{bmatrix} f(q_i,1) \\ f(q_i,2) \\ f(q_i,3) \\ \vdots \\ f(q_i,m) \end{bmatrix}.$$

It's easy to see using row reduction (take differences of successive rows) that the matrix here is in $\mathrm{GL}_m(\mathbb{Z})$, and so the equation here admits a *unique* solution, pinning down $n_1,\dots,n_m$ uniquely as desired.

(c) Conclude that the invariant factors of a finitely generated R-module are indeed uniquely determined up to invertible elements.

Solution. The previous part tells us that the elementary divisors are uniquely determined up to invertible elements. We can combine the elementary divisors to get the invariant factors using the Chinese Remainder Theorem following the same process that we used in Homework 12 of 3311 (see screenshot)

For the other direction, let $p_1, \dots, p_r$ be *distinct* primes. Let $m \geq 0$ be an integer such that we have *decreasing* sequences $a_{i,1} \geq \cdots \geq a_{i,m} \geq 0$ of integers for $i = 1, \dots, r$ such that

$$G = \prod_{\substack{1 \leq i \leq r \\ 1 \leq j \leq m}} \mathbb{Z}/p_i^{a_{i,j}}\mathbb{Z},$$

and such that there is at least one prime p_i such that $a_{i,m} \geq 1$.

Then the invariant factors for G are $d_1 \mid \cdots \mid d_m$, uniquely given by the formula

$$d_{m-j+1} = \prod_{1 \leq i \leq r} p_i^{a_{i,j}}.$$

The idea is that d_m should be the product of the highest powers of each distinct prime appearing in the elementary divisor decomposition, and then we go down the chain by taking the products of the next highest powers and so on.

Example 1. If the elementary divisors are $2^2, 2^3, 3^3, 3^4, 5^2, 13$, then $r = 4$, $p_1 = 2, p_2 = 3, p_3 = 5, p_4 = 13$ $m = 2$, and the matrix with entries $a_{i,j}$ is

$$\begin{bmatrix} 3 & 2 \\ 4 & 3 \\ 2 & 0 \\ 1 & 0 \end{bmatrix},$$

and the invariant factors are

$$d_1 = 2^2 \cdot 3^3 \text{ and } d_2 = 2^3 \cdot 3^4 \cdot 5^2 \cdot 13.$$

The idea is to take the product of the highest powers of the distinct primes appearing, and then to go down from there.

Hint: For the uniqueness, see the argument in Lecture 37 from 3311.

(2) Show that $\mathbb{Q}(\sqrt{2}, \sqrt{3}) \subset \mathbb{R}$ is Galois over $\mathbb{Q}$ and compute its Galois group.

Solution. The extension is Galois since it's a splitting field for the separable polynomial $(x^2-2)(x^2-3)$.

Within $M = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ we have the quadratic extensions $L_1 = \mathbb{Q}(\sqrt{2})$ and $L_2 = \mathbb{Q}(\sqrt{3})$.

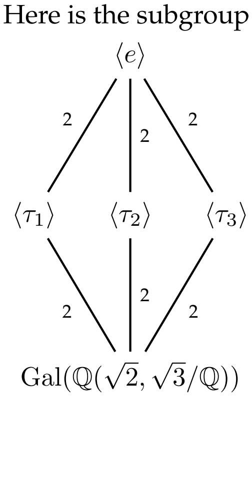
The Galois subgroup $\mathrm{Gal}(M/L_1) \leq \mathrm{Gal}(M/\mathbb{Q})$ has order 2 and is generated by an automorphism τ_1 that fixes $\sqrt{2}$ and sends $\sqrt{3}$ to $-\sqrt{3}$. This is because $M = L_1(\sqrt{3})$ is a quadratic extension of L_1.

Similarly, $\mathrm{Gal}(M/L_2)$ is generated by an automorphism τ_2 that fixes $\sqrt{3}$ and sends $\sqrt{2}$ to $-\sqrt{2}$.

Now, any automorphism of M is determined completely by what it does to $\sqrt{2}$ and $\sqrt{3}$ (why?). Therefore, one sees that $\tau_3 = \tau_1\tau_2 = \tau_2\tau_1$ is the automorphism taking $\sqrt{2}$ to $-\sqrt{2}$ and $\sqrt{3}$ to $-\sqrt{3}$.

This shows that $\mathrm{Gal}(M/\mathbb{Q}) = \{\mathrm{id}, \tau_1, \tau_2, \tau_3\}$ is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Here is the subgroup structure:



The fixed field of $\langle \tau_3 \rangle$ is a quadratic extension of $\mathbb{Q}$, and since we have

$$\tau_3(\sqrt{6}) = \tau_3(\sqrt{2}\sqrt{3}) = \tau_3(\sqrt{2})\tau_3(\sqrt{3}) = (-\sqrt{2})(-\sqrt{3}) = \sqrt{6},$$

we see that this extension must be $\mathbb{Q}(\sqrt{6})$. Therefore, the diagram of subextensions of M looks like

$$\begin{tikzcd}
 & \mathbb{Q}(\sqrt{2},\sqrt{3}) \arrow[dl, dash, "2"'] \arrow[d, dash, "2"] \arrow[dr, dash, "2"] & \\
\mathbb{Q}(\sqrt{2}) \arrow[dr, dash, "2"'] & \mathbb{Q}(\sqrt{3}) \arrow[d, dash, "2"] & \mathbb{Q}(\sqrt{6}) \arrow[dl, dash, "2"] \\
 & \mathbb{Q} &
\end{tikzcd}$$

Let $\zeta_n = e^{2\pi i/n} \in \mathbb{C}$ be a *primitive* n-th root of 1. Define

$$\Phi_n(x) = \prod_{j \in (\mathbb{Z}/n\mathbb{Z})^\times} (x - \zeta_n^j) \in \mathbb{C}[x].$$

This is the **n-th cyclotomic polynomial**.

(3) When $n = p$ is prime, show that $\Phi_p(x) = x^{p-1} + \cdots + x + 1$, and use Eisenstein's criterion to conclude that $m_{\zeta_p}(x) = \Phi_p(x) \in \mathbb{Q}[x]$.

Solution. In $\mathbb{C}[x]$, we have

$$x^p - 1 = \prod_{j=0}^{p-1} (x - \zeta_p^j),$$

and so we conclude that we have

$$\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + \cdots + x + 1 \in \mathbb{Z}[x].$$

To use Eisenstein's criterion, we make the substitution $y = x - 1 \Leftrightarrow x = y + 1$. This gives us

$$\frac{(y+1)^p - 1}{y} = y^{p-1} + \sum_{j=1}^{p-2} \binom{p}{j} y^{p-j-1} + p.$$

We can see that this is irreducible by using Eisenstein's criterion with the prime p. Since $\Phi_p(\zeta_p) = 0$, we conclude that $m_{\zeta_p}(x) = \Phi_p(x)$.

(4) Show that $\mathbb{Q}(\zeta_p) \subset \mathbb{C}$ is Galois over $\mathbb{Q}$ with Galois group isomorphic to $(\mathbb{Z}/p\mathbb{Z})^\times$.

Solution. It's easy to see that $\mathbb{Q}(\zeta_p)$ is a splitting field for the separable polynomial $\Phi_p(x)$, and is therefore Galois over $\mathbb{Q}$. As usual, we have a bijection

$$\mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) = \mathrm{Emb}_{\mathbb{Q}}(\mathbb{Q}(\zeta_p), \mathbb{Q}(\zeta_p)) \simeq \{\alpha \in \mathbb{Q}(\zeta_p) : \ \Phi_p(\alpha) = 0\} = \{\zeta_p^i : i \in (\mathbb{Z}/p\mathbb{Z})^\times\}.$$

Concretely, this carries each $\sigma \in \mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$ to $\sigma(\zeta_p)$.

For each $i \in (\mathbb{Z}/p\mathbb{Z})^\times$, let $\sigma_i \in \mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$ be the automorphism satisfying $\sigma_i(\zeta_p) = \zeta_p^i$. The function $i \mapsto \sigma_i$ gives a bijection $(\mathbb{Z}/p\mathbb{Z})^\times \to \mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$. It is actually an isomorphism, since

$$(\sigma_i \circ \sigma_j)(\zeta_p) = \sigma_i(\zeta_p^j) = \sigma_i(\zeta_p)^j = (\zeta_p^i)^j = \zeta_p^{ij} = \sigma_{ij}(\zeta_p).$$

(5) Suppose that we have a Galois extension L/k, and that K/k is a k-sub-extension of L. Given $\sigma \in \mathrm{Gal}(L/k)$, show that

$$\mathrm{Gal}(L/\sigma(K)) = \sigma\,\mathrm{Gal}(L/K)\sigma^{-1}.$$

Conclude that we have $\sigma(K) = K$ for all $\sigma \in \mathrm{Gal}(L/k)$ if and only if $\mathrm{Gal}(L/K) \trianglelefteq \mathrm{Gal}(L/k)$ is a normal subgroup.

Hint: We will see in class that L is Galois over K, so that $\mathrm{Gal}(L/K)$ makes sense. Think of $\mathrm{Gal}(L/K)$ as the simultaneous stabilizer in $\mathrm{Gal}(L/k)$ of all the elements in K.

Solution. This is quite standard: In any group action the stabilizer $G_{g\cdot x}$ is equal to gG_xg^{-1}. Concretely, $h \cdot x = x$ implies that $(ghg^{-1}) \cdot (g \cdot x) = g(hg^{-1}g \cdot x) = g \cdot x$.

The automorphisms of L over k acting trivially on K can be seen as the simultaneous stabilizer of all the elements of K, and so the subgroup acting trivially on $\sigma(K)$ is the conjugate by σ, as desired.

If $\sigma(K) = K$, then this tells us that $\sigma\,\mathrm{Gal}(L/K)\sigma^{-1} = \mathrm{Gal}(L/K)$, and so if this holds for all σ, then we conclude that $\mathrm{Gal}(L/K) \trianglelefteq \mathrm{Gal}(L/k)$ is a normal subgroup. The other direction was shown in Proposition 2 of Lecture 36.

(6) Suppose that p is odd, and let $H \leq (\mathbb{Z}/p\mathbb{Z})^\times$ be the unique subgroup of order 2. Compute the fixed field

$$\mathbb{Q}(\zeta_p)^H = \{\alpha \in \mathbb{Q}(\zeta_p) :\ \sigma(\alpha) = \alpha \text{ for all } \sigma \in H\}.$$

Solution.

(7) Show that $\mathbb{Q}(\sqrt[4]{2}, i)$ is Galois over $\mathbb{Q}$, and compute its Galois group. If you're feeling ambitious, write down all of its subextensions as well, or at least figure out how many such subextensions there are.

Solution. Set $M = \mathbb{Q}(\sqrt[4]{2}, i)$. Note that we have a sequence of extensions

$$\mathbb{Q} \leq \mathbb{Q}(\sqrt[4]{2}) \leq M,$$

where the first extension has degree 4 (the minimal polynomial is $x^4 - 2$ by Eisenstein's criterion), and the second extension has degree 2 since it is obtained by adjoining the imaginary number i which satisfies a quadratic polynomial (note that $i \notin \mathbb{Q}(\sqrt[4]{2})$, because all the elements of this field have to be real numbers).

For the description of the Galois group, we basically follow the method we used in class for $\mathbb{Q}(\sqrt[3]{2}, \zeta_3)$.

We see that $[M : \mathbb{Q}] = 8$. Inside of M we have the subextensions:

$$L_1 = \mathbb{Q}(\sqrt[4]{2})\ ;\ L_2 = \mathbb{Q}(i).$$

with $[M : L_1] = 2$ and $[M : L_2] = 4$. This means that we have subgroups

$$\mathrm{Gal}(M/L_1), \mathrm{Gal}(M/L_2) \leq \mathrm{Gal}(M/\mathbb{Q})$$

of orders 2 and 4, respectively.

Now, we have $M = \mathbb{Q}(i)(\sqrt[4]{2}) = L_2(\sqrt[4]{2})$, and the minimal polynomial of $\sqrt[4]{2}$ over L_3 is still $x^4 - 2$ (why?). This means that there is a bijection

$$\mathrm{Gal}(M/L_2) = \mathrm{Emb}_{L_2}(M, M) \xrightarrow{\simeq} \{\pm\sqrt[4]{2}, \pm i\sqrt[4]{2}\}.$$

So we have four elements $\{\mathrm{id}, \sigma_{-1}, \sigma_i, \sigma_{-i}\}$ in $\mathrm{Gal}(M/L_2)$, where id is the identity and $\sigma_\alpha(\sqrt[4]{2}) = \alpha\sqrt[4]{2}$ for $\alpha = \pm1, \pm i$. All of these automorphisms fix $\pm i \in L_2$. Set $\sigma = \sigma_i$: then we have $\sigma_{-1} = \sigma^2$, $\sigma_{-i} = \sigma^3$ and of course $\sigma^4 = \mathrm{id}$. In other words, $\mathrm{Gal}(M/L_2)$ is cyclic of order 4 generated by σ.

What about $\mathrm{Gal}(M/L_1)$? Since $M = L_1(i)$, we should have

$$\mathrm{Gal}(M/L_1) = \mathrm{Emb}_{L_1}(M, M) \xrightarrow[\tau\mapsto\tau(i)]{\simeq} \{\pm i\}.$$

This means that $\mathrm{Gal}(M/L_1)$ has two elements τ_1, τ_{-1}, both of which fix $\sqrt[4]{2} \in L_1$, and which satisfy $\tau_j(i) = i^j$, where $j = \pm1$ (note that $i^{-1} = -i$). So $\mathrm{Gal}(M/L_1)$ is generated by $\tau = \tau_{-1}$.

Now, any automorphism of M over $\mathbb{Q}$ is clearly determined by what it does to $\sqrt[4]{2}$ and i.

Given our experience with $\sqrt[3]{2}$, we might guess that the Galois group should be D_8, generated by σ and τ in the usual way. For this, we have to check the relation $\sigma\tau = \tau\sigma^3$. We compute:

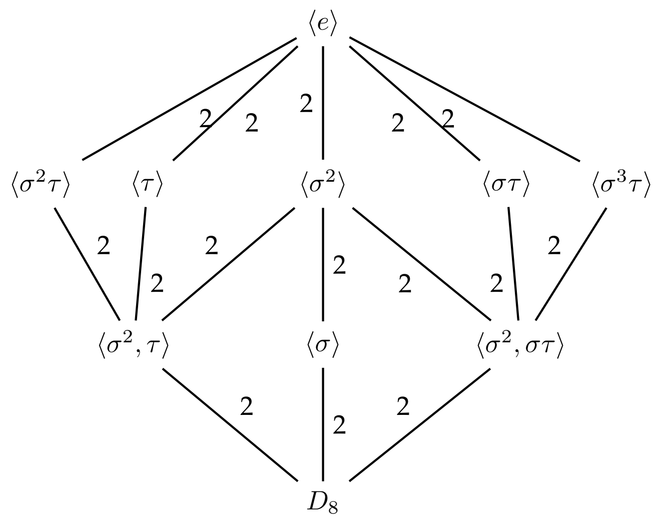
$$(\sigma\tau)(i) = \sigma(-i) = -i$$
$$(\sigma\tau)(\sqrt[4]{2}) = \sigma(\sqrt[4]{2}) = i\sqrt[4]{2}.$$

Also,

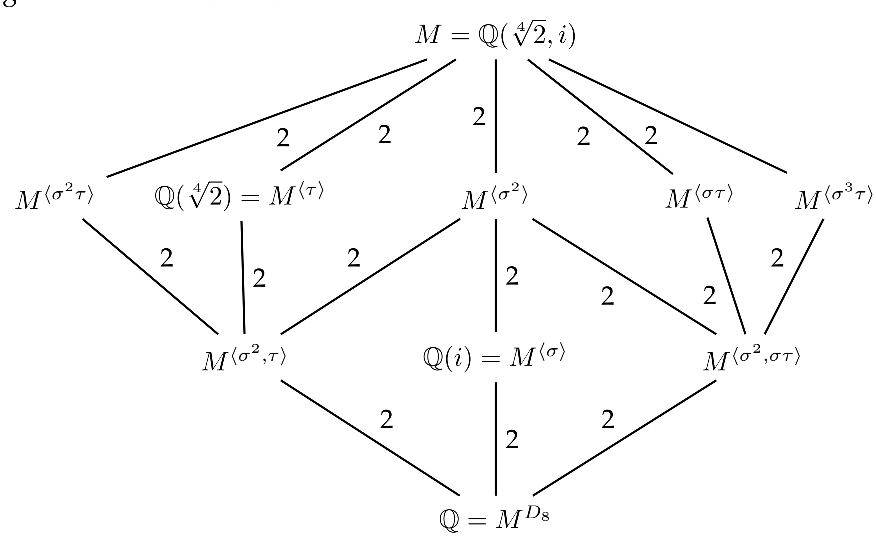
$$(\tau\sigma^3)(i) = \tau(i) = -i$$
$$(\tau\sigma^3)(\sqrt[4]{2}) = \tau(-i\sqrt[4]{2}) = \tau(-i)\sqrt[4]{2} = i\sqrt[4]{2}.$$

This tells us that $\sigma\tau = \tau\sigma^3$ as desired.

To explore the structure of subextensions, let us write down the diagram of subgroups of D_8. The labels on the connecting lines are the index of the smaller subgroup in the larger.



We'd like to compute the corresponding diagram of subextensions of M. The labels are now the degree of each field extension.



Here, we have used the fact that we already know that σ (resp. τ) generates the Galois group $\mathrm{Gal}(M/\mathbb{Q}(i))$ (resp. $\mathrm{Gal}(M/\mathbb{Q}(\sqrt[4]{2}))$), and so we get the matching via the correspondence of Galois theory.

To determine the rest, note first that $M^{\langle \sigma^2, \tau \rangle}$ and $M^{\langle \sigma^2, \sigma\tau \rangle}$ should both be quadratic extensions of $\mathbb{Q}$. But we can already write down some quadratic subextensions of M: Namely $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(i\sqrt{2}) = \mathbb{Q}(\sqrt{-2})$.

We have $\sigma(\sqrt{2}) = \sigma((\sqrt[4]{2})^2) = (i\sqrt[4]{2})^2 = -\sqrt{2}$. This shows that $\sigma^2(\sqrt{2}) = \sqrt{2}$. Similarly,

$$\sigma(\sqrt{-2}) = \sigma(i\sqrt{2}) = \sigma(i)\sigma(\sqrt{2}) = i(-\sqrt{2}) = -\sqrt{-2}.$$

Therefore, $\sigma^2(\sqrt{-2}) = \sqrt{-2}$. Note that this shows that we have the degree 4 extension

$$\mathbb{Q}(\sqrt{2}, \sqrt{-2}) = \mathbb{Q}(\sqrt{2}, i) \leq M^{\langle \sigma^2 \rangle}.$$

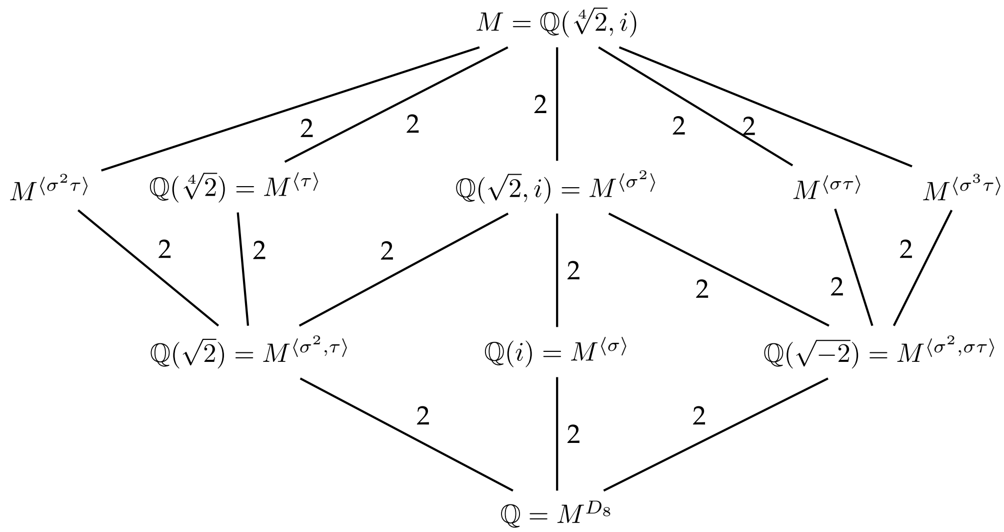
By degree considerations, this is actually an equality.

Now, $\tau(\sqrt[4]{2}) = \sqrt[4]{2}$ and hence $\tau(\sqrt{2}) = \sqrt{2}$. This shows that we have

$$\mathbb{Q}(\sqrt{2}) \leq M^{\langle \sigma^2, \tau \rangle},$$

and this is an equality by degree considerations. We are therefore forced to have $\mathbb{Q}(\sqrt{-2}) = M^{\langle \sigma^2, \sigma\tau \rangle}$.

Let us collect what we have so far:



Now, let us make the following observation:

$$\sigma\tau\sigma^{-1} = \sigma\sigma\tau = \sigma^2\tau \ ; \ \sigma(\sigma\tau)\sigma^{-1} = \sigma^2\sigma\tau = \sigma^3\tau$$

Therefore, if $H = \langle \tau \rangle = \mathrm{Gal}(M/L_1))$, then $\sigma H \sigma^{-1} = \langle \sigma^2\tau \rangle$, but this is $\mathrm{Gal}(M/\sigma(L_1)) = \mathrm{Gal}(M/\mathbb{Q}(i\sqrt[4]{2}))$ by problem 5.

Similarly, if $H' = \langle \sigma\tau \rangle = \mathrm{Gal}(M/L_1')$, then $\langle \sigma^3\tau \rangle = \mathrm{Gal}(M/\sigma(L_1'))$. To finish, we have to know what $L_1' = M^{\langle \sigma\tau \rangle}$ is.

A reasonable guess is that we should have $L_1' = \mathbb{Q}(\sqrt[4]{-2})$ (for some good definition of $\sqrt[4]{-2}$). Is this actually a subextension of M? Yes! It's because we have square-roots of $\pm i$ given by $\pm\frac{1 \pm i}{\sqrt{2}}$, and these do belong to M and in fact to $M^{\langle \sigma^2 \rangle}$. Therefore, the elements

$$\left(\pm \frac{1 \pm i}{\sqrt{2}} \right) \cdot \sqrt[4]{2}.$$

give us fourth roots of -2.

Now, note that

$$\sigma\left(\frac{1+i}{\sqrt{2}} \sqrt[4]{2} \right) = \frac{\sigma(1) + \sigma(i)}{\sigma(\sqrt{2})} \sigma(\sqrt[4]{2}) = \frac{1+i}{-\sqrt{2}} (i\sqrt[4]{2}) = \frac{1-i}{\sqrt{2}} \sqrt[4]{2}.$$

Also,

$$\tau\left(\frac{1+i}{\sqrt{2}} \sqrt[4]{2} \right) = \frac{\tau(1) + \tau(i)}{\tau(\sqrt{2})} \tau(\sqrt[4]{2}) = \frac{1-i}{\sqrt{2}} \sqrt[4]{2}$$

These two computations show that $\sigma^{-1}\tau = \sigma^3\tau$ fixes $\frac{1+i}{\sqrt{2}}\sqrt[4]{2}$, and so we conclude that

$$M^{\langle \sigma^3\tau\rangle} = \sigma(L_1') = \mathbb{Q}\left(\frac{1+i}{\sqrt{2}}\sqrt[4]{2}\right)$$

and, since

$$\sigma^{-1}\left(\frac{1+i}{\sqrt{2}}\sqrt[4]{2}\right) = -\frac{1-i}{\sqrt{2}}\sqrt[4]{2}$$

we conclude that

$$L_1' = \sigma^{-1}(\sigma(L_1')) = \mathbb{Q}\left(\frac{1-i}{\sqrt{2}}\sqrt[4]{2}\right).$$

Therefore, the complete picture is the following: